Weekly Business Review

Olist E-commerce Marketplace | Data Quality, Commercial Performance & Delivery

99,441 orders | 91.89% on-time delivery | 4.09 / 5 average review

1. Executive Summary

The analysis mart contains **99,441 rows and 99,441 distinct order IDs**. The on-time delivery rate is **91.89%**, the late-delivery rate is **8.11%**, the average review score is **4.09/5**, and average order value is **159.33** in the dataset's currency, including product price and freight.

The North Star metric is **on-time delivery rate** because it measures whether delivered orders arrive by the estimated date. Review scores add a customer-experience measure, while late-delivery rate acts as a direct operational guardrail.

2. Pipeline and Methodology

Raw ingestion: CSV files are loaded into PostgreSQL raw tables. **Staging:** relevant fields and data types are prepared. **Analysis mart:** item, payment, and review data are aggregated before joining to orders to reduce the risk of multiplying order-level records. **Analysis:** SQL calculates monthly order value, state-level category rankings, rolling order counts, repeat purchasing, and delivery/review measures.

The pipeline log records successful execution. The mart contains 99,441 rows and 99,441 unique orders, so the row count agrees with the distinct order count. The results describe the supplied Olist dataset and should not be interpreted as current marketplace performance.

Business question

Where should leadership focus to improve customer satisfaction without losing sight of order value, repeat purchasing, and data reliability?

3. Data Quality Results

Nine automated checks were executed. Six critical checks returned zero failures. Three additional checks identified exceptions that were flagged for investigation rather than silently discarded. In the current pipeline, these three checks are warnings and do not stop mart creation.

Check	Failed rows	Handling / interpretation
Missing order IDs	0	Passed
Duplicate order IDs	0	Passed
Invalid delivery dates	0	Passed
Invalid item values	0	Passed
Missing customer records	0	Passed
Invalid review scores	0	Passed
Payment mismatches	267	Flag for investigation
Orders without items	775	Review order status; do not silently delete
Orders without payments	1	Flag the unmatched delivered order

What the exceptions mean

The 267 payment mismatches are cases where aggregated item-plus-freight totals differ from payment totals by more than 0.02. The 775 orders without items are largely associated with unavailable or canceled statuses. One delivered order has no matching payment record. These exceptions are not, by themselves, evidence of fraud; investigate source records before changing them.

Control note: the pipeline currently logs these three checks as warnings and continues. The assignment asks for an error/alarm when a check fails, so strict fail-fast behavior for these checks remains a follow-up improvement.

4. Supporting Business Metrics

Metric	Result	Formula / definition
North Star: on-time delivery	91.89%	Delivered orders on/before estimate ÷ delivered orders with known outcome
Late-delivery rate	8.11%	Delivered orders after estimate ÷ delivered orders with known outcome
Average review score	4.09 / 5	Mean available review score
Average order value	159.33	Mean (item price + freight) per order
Repeat purchase rate	3.12%	Unique customers with >1 order ÷ all unique customers
Repeat-customer order share	6.38%	Orders placed by repeat customers ÷ all orders

Interpretation: repeat purchase rate is customer-based, whereas repeat-customer order share is order-based; they use different denominators.

5. Commercial Performance

The charts below summarize revenue/order-value movement and product-category concentration. Interpret changes alongside order counts, especially at the end of the dataset period.

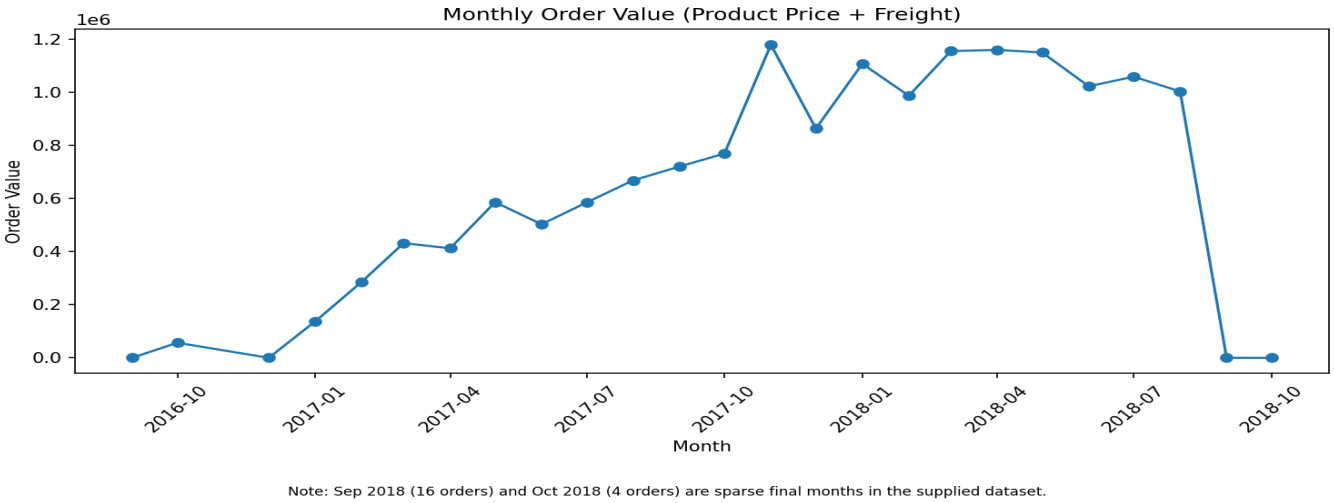


Figure 1. Monthly order value and month-over-month growth. Sep 2018 (16 orders) and Oct 2018 (4 orders) are sparse final months in the supplied dataset.

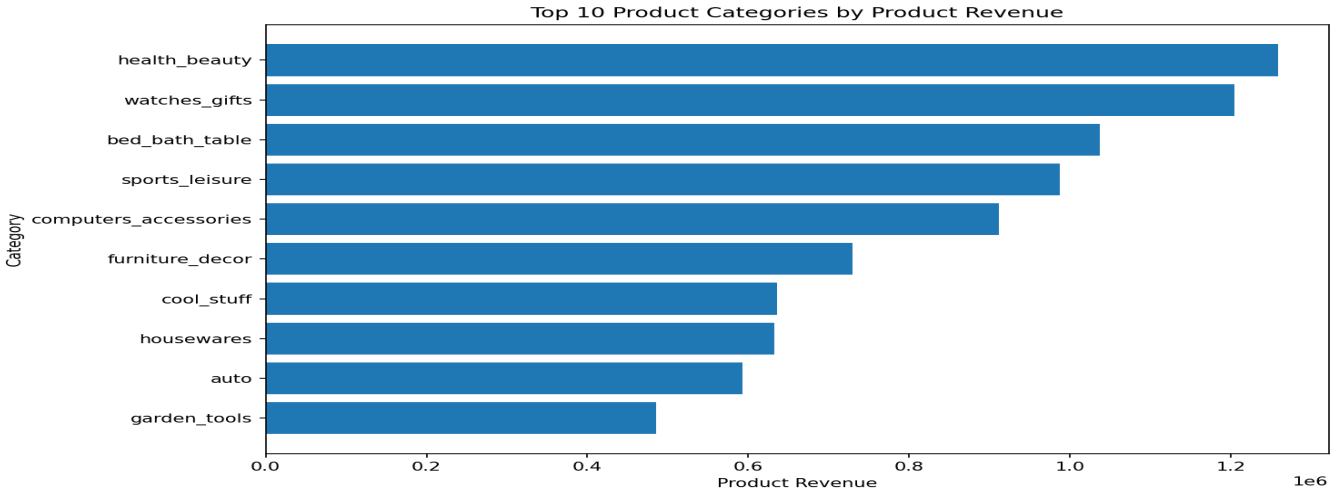


Figure 2. Top product categories by customer state, ranked by revenue.

How to use these results

Use monthly movement to identify periods that merit follow-up, but avoid over-interpreting the final sparse months. State-level category rankings can help identify where demand is concentrated and where assortment, availability, or seller coverage may deserve closer review.

6. Orders and Customer Experience

Order volume and delivery experience should be monitored together: volume shows demand patterns, while delivery outcomes and review scores help reveal potential friction in the customer journey.

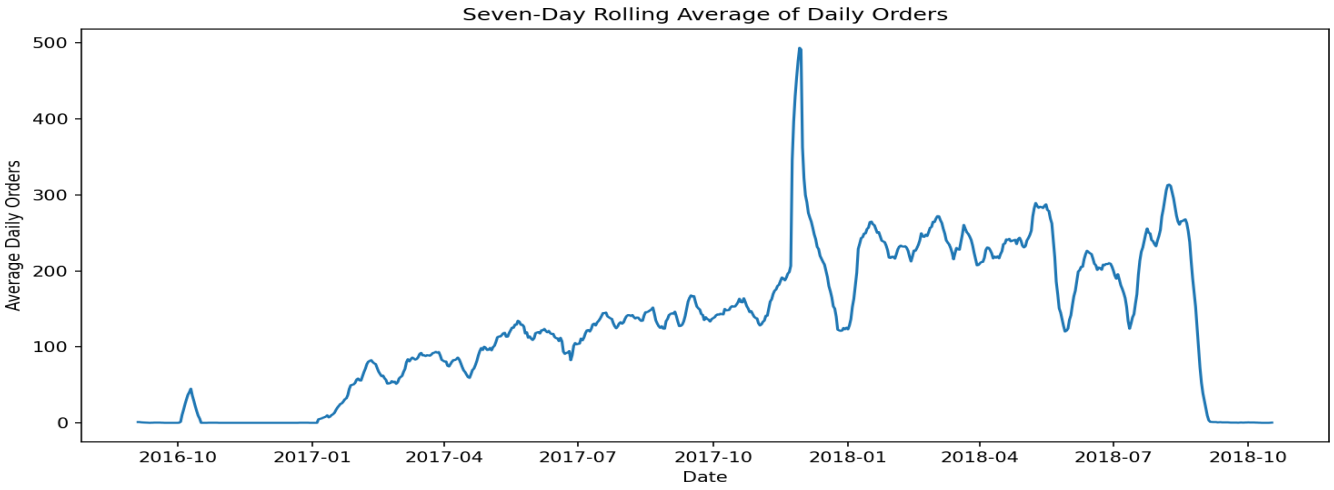


Figure 3. Seven-day rolling average of daily orders, including calendar days with zero orders.

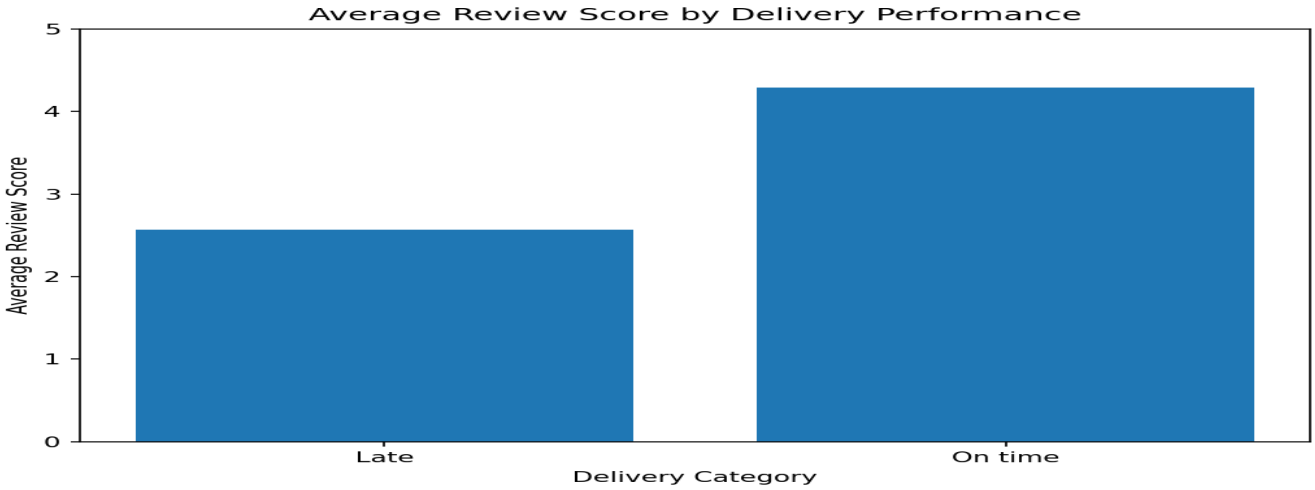


Figure 4. Average review score by delivery performance.

Late deliveries average 2.57/5 in review score, compared with 4.29/5 for early deliveries. This is an association in the observed data, not proof that delivery timing alone caused the review difference.

Operational follow-up should compare late-delivery patterns by customer state, seller, category, and time period. Track whether targeted changes are followed by improvements in both on-time delivery and review scores.

7. VP Memo: Improving Customer Experience

To: Vice President, Marketplace Operations **Subject:** Delivery reliability as a customer-experience priority

Key finding

The dataset shows an overall on-time delivery rate of 91.89%, with 8.11% of delivered orders arriving late. Late deliveries are associated with a lower average review score: 2.57/5 for late deliveries versus 4.29/5 for early deliveries. This gap makes delivery reliability a practical priority for investigation, while not proving causation.

Recommended action

Run a focused late-delivery review by customer state, seller, product category, and purchase period. Identify the segments contributing the largest number of late orders, validate likely causes with operational teams, and pilot a targeted corrective action. Compare on-time delivery and review scores before and after the pilot.

Weekly scorecard

Measure	Why leadership should monitor it
On-time delivery rate (North Star)	Primary indicator of delivery reliability
Late-delivery rate	Guardrail that highlights failed delivery commitments
Average review score	Customer-experience signal to read alongside delivery
Average order value	Tracks order economics; includes item price and freight
Repeat purchase rate	Tracks the share of customers who return to order

Data and decision caveats

Payment mismatches (267), orders without items (775), and orders without payments (1) remain flagged for investigation. Resolve or document these exceptions and strengthen the pipeline's warning/failure policy before using affected records for operational or financial decisions. The Olist data spans 2016–2018; findings are historical and may not reflect current performance.

Data source: Olist Brazilian E-Commerce Public Dataset. **Tools:** Python, PostgreSQL, SQL, pandas, and matplotlib.